

BismillahirRahmanir Raheem

International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering

Final Examination, Autumn- 2025

Course Code: CHEM-2301

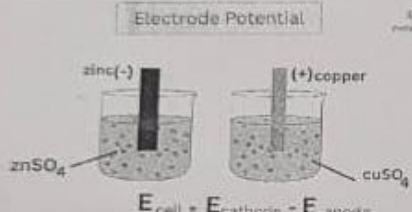
Course Title: Chemistry

Total Marks: 50

Time: 2 Hour 30 Min

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Separate answer scripts must be used for Group A and B.
(iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Group A

1.	(a)	State Electrolysis and categorize it. Evaluate Faraday's 1st law of electrolysis with its significance.	E	CLO1	5
	(b)	Define Transference number and ionic mobility. The speed ratio of silver and nitrate ions in an aqueous solution of silver nitrate is 0.916. Calculate the transport number of both ions.	An	CLO2	5
2.	(a)	Electrode Potential  $E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$ Cite electrode potential. Briefly describe about the procedures of above electrode potentials system.	E	CLO3	5
	(b)	Discuss about molarity, molality, and normality. Distinguished between saturated, unsaturated and supersaturated solutions.	An	CLO2	5
		OR			
	(a)	Explain Raoult's Law. Distinguish between ideal and real solutions.	R	CLO1	5
	(b)	State Nernst's Distribution Law and mention the colligative properties of dilute solutions.	An	CLO2	5

Group B

3.	(a)	What is reversible reaction. "Chemical equilibrium is a dynamic process"- Explain.	Ap	CLO3	5
	(b)	Mention Le Chatelier's Principle. How do you get the more profit for the synthesis of ammonia by Haber-Bosch process? Explain according to Le Chatelier's Principle.	U	CLO3	5
4.	(a)	Express reaction rate law, "A→Product" Determine the reaction order equation for this reaction.	E	CLO1	5
	(b)	Graphically represent the 1st order reaction curve. Show the relationship between activation energy and Arrhenius equation.	An	CLO1	5
5.	(a)	Define adsorption and explain the differences between physisorption and chemisorption with suitable examples.	C	CLO3	5
	(b)	Discuss the factors affecting adsorption on solid surfaces with proper examples.	C	CLO2	5
		OR			
	(a)	Define colloids and mention their classification based on physical state, and interaction with examples.	E	CLO1	5
	(b)	Mention the preparation methods of colloids. Explain any method in details.	Ap	CLO3	5

max amount of solute

International Islamic University Chittagong

Department of Computer Science and Engineering

Final Examination, Autumn 2025

Course Code: CSE 2323

Total Marks: 50

Course Title: Digital Logic Design

Time: 2 hours 30 minutes

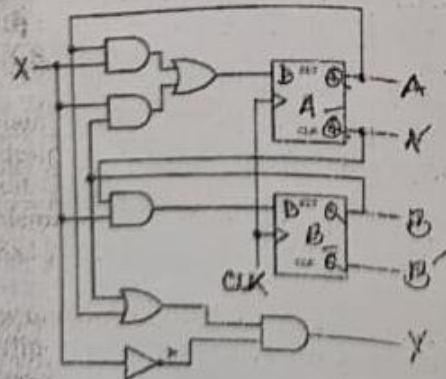
- The figures in the right-hand margin indicate full marks.
- Course Outcomes and Bloom's Levels are mentioned in additional columns.
- Separate answer scripts must be used for Group A and Group B.

Group-A

- Design a 4-to-14 decoder by using two 2-to-4 decoders. For the input combination $A_3 A_2 A_1 A_0 = 1101$, determine which output line becomes active. CO2 5
 - Suppose you need to design a 1-bit full adder, but you only have two 4-to-1 multiplexers. Show how the full adder can be implemented using two 4-to-1 MUX. Specify select lines and data inputs. CO2 5
- OR**

 - Explain how an 8×1 multiplexer can be constructed using 4-to-1 MUX. Illustrate the design with a neat diagram and describe the working principle of the resulting circuit. CO2 5
 - Implement the Boolean function $F(A, B, C, D) = \sum(1, 3, 5, 7, 9, 10, 12, 15)$ using an 8×1 MUX. CO2 5
 - How does JK flip-flop remove the indeterminate states of S-R flip-flop? Design a JK flip-flop and show its characteristic equation, characteristic table, logic diagram. CO3 5
- Or**

 - The following circuit consists of two D flip-flops, A and B, an input 'x', and an output 'y'.
 - Derive the next-state equations for flip-flops A and B, and the output Boolean function for 'y'.
 - Construct the state table for this sequential circuit.
 Draw the corresponding state diagram. CO2 5



Group-B

3.	a)	Using JK flip-flop, design a synchronous counter that produces the following sequence: 0, 1, 2, 3, 5, 7, and then repeats. Draw the state diagram, excitation table, and circuit diagram.	CO2	5
	b)	Design a 4-bit asynchronous Up counter with JK flip-flops and explains its operation with timing diagram.	CO3	5
	Or			5
	b)	A device receives 4-bit parallel data from multiple sensors and needs to send it out bit by bit over a single communication line. Explain with diagram how the device can handle this data conversion.	CO3	5
4.	a)	1. Design and simulate a parallel adder using full adder to add the binary numbers $A = 1101$ and $B = 1101$ with an initial Carry-In = 1. Show the sum and carry generated at each bit position, including the final output. 2. Analyze the effect of the initial carry-in on the adder's operation. Explain how it influences carry propagation and the final result.	CO2	5
	b)	Implement the following Boolean functions using a Programmable Logic Array (PLA) $f = \sum m(0, 1, 7)$ $g = \sum m(0, 3, 6, 7)$ $h = \sum m(0, 1, 3, 5, 7)$	CO3	5
	Or			
	b)	A device receives 4-bit parallel data from multiple sensors and needs to send it out bit by bit over a single communication line. Explain with diagram how the device can handle this data conversion.	CO3	5
5.	a)	Design a <i>Serial In Parallel Out</i> shift register and derive its state transition behavior.	CO1	5
	b)	Design a 4x5 RAM and describe its operation with example. Analyze how many address lines are required to uniquely select a memory location in a 4×5 RAM and justify your answer.	CO2	5

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in Computer Science and Engineering

Semester Final Examination, Autumn 2025

Course Title: Data Structures

Course Code: CSE-2321

Time: 2 Hours 30 Minutes

Full Marks: 50

[Answer all the questions. Figures in the right margin indicate full marks. Separate answer scripts must be used for Group A and Group B]

[Answer all the questions. Figure in the right margin indicate full marks. Separate answer scripts must be submitted for each question.]				DL	CO	Marks
Group A						
1	a)	Explain two basic operations performed on a stack. List two applications of stacks in computer science.		C3	CO1	3
OR						
	a)	Define a stack . Explain how a stack can be represented using an array .				
	b)	Suppose the following stack of integers is in memory where STACK is allocated N=8 memory cells: TOP=3 STACK: 5,2,3,_,_,_,_,_ (_ denotes an empty cell) Find the output of the following module: 1. Call POP(STACK, ITEM). Call POP(STACK, ITEM). Call PUSH(STACK, ITEM+N). Call PUSH(STACK, N). Call PUSH(STACK, ITEM+ITEM). 2. Repeat while TOP != 0: Call POP(STACK, ITEM). Write: ITEM. [End of loop] Return. Note: N is the sum of the digits of your ID, i.e., ID=C123024, N=1+2+3+0+2+4=12.	C3	CO2	4	
	c)	Evaluate the following postfix expression using the algorithm that you have studied. 3, 1, +, 2, ^, N, 4, -, 2, *, +, 5, - Note: N is the sum of the digits of your ID, i.e., ID=C123024, N=1+2+3+0+2+4=12.	C3	CO2	3	
OR						
	c)	Translate the following infix expressions into its equivalent post-fix expression using the algorithm that you have studied. $A*(B+D)/E-F*(G+H/K)$				
2	a)	What is a queue ? Are there any limitations of a queue? If yes, explain them and how they can be overcome.		C2	CO1	3
OR						
	a)	What is priority queue ? Discuss the <i>one way list</i> representation of a priority queue. Compare the <i>one way list</i> and <i>array</i> representation of priority queues.				
	b)	Consider the following queue of characters where QUEUE is a circular array which is allocated 5 memory cells: FRONT = 2, REAR = 4 QUEUE: _, A, B, C, _ Describe the queue, including FRONT and REAR, as the following operations take place – 1. P is added to the queue. 2. Two letters are deleted. 3. M, N, O are added to the queue. 4. Three letters are deleted.	C3	CO2	3	
	c)	A recursive function is defined as follows: $T(n) = 1, \text{ if } n = 0$ $T(n) = n \times T(n-1), \text{ if } n > 0$ (i) Identify the mathematical function represented by T(n). (ii) Compute the value of T(4).	C3	CO2	2	
OR						
	c)	A function F(n) is defined recursively as follows: $F(n) = 0, \text{ if } n = 0$ $F(n) = F(n/2) + 1, \text{ if } n > 0$ Find the value of F(8).				

International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination, Autumn-2025
Course Code: GEED-2302 (GEED-2101 for LLB)
Course Title: Sciences of Qur'an and Hadith

Full Marks: 50

Time: 2 Hours 30 Minutes

(Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question):

Questions	Marks	CLOs	Bloom's taxonomy domain
1/ a. Define Naskh (abrogation). State some benefits of Naskh (abrogation) of the holy Qur'an.	5/	3	Remember
b. Categorize some classifications of Naskh (abrogation) with examples.	5		Understand
Or,			
c. Discuss the characteristics of Makki and Madani revelations.	5/		Remember
d. Describe some opinions of Muslim scholars regarding the benefits of Makki and Madani revelations.	5	3	Understand
2/ Relate the history of compilation of the holy Qur'an (Jam'ul Qur'an) during the caliphate of Abu Bakr (ra) and Uthman (ra).	10/	3	Apply
3 a. Critically evaluate the concept of I'jāz al-Qur'an selecting what is the <u>best miracle</u> of the Prophet (SAAS).	5/	3	Evaluate
b. "At whatever angle we look the holy Qur'an it is Mu'jiza (Miracle)"- justify.	5/		Create
4/ Relate the concept of Hadith literally and terminologically. Point out the history of compilation, and preservation of Hadith.	10/	3	Analyze
5/ Explain with examples: (Any four)			
a. Al-Sanad and Al-Matan;			
b. Al-Hadith Al-Mutawatir;			
c. Al-Hadith Al-Marfu';	10/	3	Create
d. Al-Hadith Al-Mauquf;			
e. , Al-Hadith Al-Maqtu'.			

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International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Semester Final Examination

Program: B. Sc. in CSE
 Course Code: MATH-2307
 Time: 2.5 hours



Semester: Autumn 2025
 Course Title: Mathematics-III
 Total Marks: 50

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
- (ii) Please answer the several parts of a question sequentially.
- (iii) Separate answer script must be used for separate group.
- (iv) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1:	Understand the fundamentals of Matrix, Linear system of equations & Vector analysis.
CLO2:	Implement the fundamental knowledge of Matrix, linear system of equations, vector functions, vector field, scalar field, gradient, divergence, curl, differentiation and integration of vector valued functions, partial derivatives in different problems.
CLO3:	Solve line integrals, surface area, surface integrals, volume integrals, and the work done in different problems.
CLO4:	Apply Green's theorem, Stoke's theorem, and Gauss' theorem in solving mathematical problems.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Part A

1. Examine the Eigen decomposition for the matrix $\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ CLO2 An 10

2. a) Find a unit vector parallel to the resultant of vectors, $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{B} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ CLO1 U 5

Or

- If $\phi(x, y, z) = xy^2z$ and $\vec{A} = xz\hat{i} - xy^2\hat{j} + yz^2\hat{k}$ then find $\frac{\delta^2}{\delta x^2 \delta z}(\phi \cdot \vec{A})$ at the point (2, -1, 1) CLO1 U 5

2. b) A particle moves through 3-space in such a way that its velocity is $\vec{v}(t) = \hat{i} + t\hat{j} + t^2\hat{k}$. Find the co-ordinates of the particle at time $t = 1$ given that the particle is at the point (-1, 2, 4) at time $t = 0$ CLO2 U 5

Or

Find all vectors $\vec{v}$ such that $(\hat{i} + 2\hat{j} + \hat{k}) \times \vec{v} = 3\hat{i} + \hat{j} - 5\hat{k}$

CLO2 U 5

Part B

3. a) Show that $\vec{\nabla} \phi$ is a vector perpendicular to the surface $\phi(x, y, z) = c$, where c is a constant

CLO2 U 5

b) Find the directional derivative of the function $\phi = x^2z + 2xy^2 + yz^2$ at the point of $(1, 2, -1)$ in the direction of the vector $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$.

CLO2 U 5

4. a) If $\vec{F} = 2xy\hat{i} - y^2\hat{j}$ then evaluate $\int_c \vec{F} \cdot d\vec{r}$, where c is the curve in the xy plane, $y = 2x^2$ from $(0, 0)$ to $(1, 2)$

CLO3 U 5

b) Determine the area enclosed by the graphs of $y = x^3$ and $y = 4x$ for $x \geq 0$

CLO3 U 5

5. Verify the divergence theorem for the vector field

CLO4 App 10

$\vec{F} = x^2\hat{i} + z\hat{j} + y\hat{k}$ taken over the region bounded by the planes $z = 0, z = 2, x = 0, x = 1, y = 0, y = 3$

Or

Verify Green's theorem in the plane for $\oint_c \{(xy + x^2) dx + xy^2 dy\}$ where the close curve of the region bounded by $y = x$ and $y^2 = x$

CLO4 App 10

International Islamic University Chittagong
Morality Development Program (MDP)
Semester End Examination, Autumn- 2025
Course Title: Tazkiyahtun Naf's, Course Code: MDP-2302A,
3rd Semester

Full Marks : 50

Time: 2.5 Hours

[Part [A] Answer each of the following question: 3 x 10 = 30

Answer the following questions

Q. [1] Write down the complete sentence choosing the correct option:

i. Good deed is considered:

- a) The main principle on which Tahliya is founded
- b) One of the basic elements of ornamentation of soul
- c) The only element on which Tahliya is founded.

ii. To attain Ikhlas (sincerity of intention) one must:

- a) Perform each Ibadah with excellence
- b) Purify his intention and do this for the cause of Allah
- c) To be conscious about the criticism of others.

iii. Paying Zakat is considered as:

- a) The main source for attaining Tazkiya.
- b) An additional quality for attaining Tazkiyah.
- c) One of the essential sources for attaining Tazkiyah.

iv. Kindness and compassion mean being gentle and caring:

- a) Towards everyone regardless of friends and foes
- b) Towards the family members and fellows
- c) Towards the foes and critics.

v. Studying the Seerah of the Prophet (SAW) is:

- a) Closely related to purification of soul (Tazkiyah)
- b) Closely related to adorning the soul (Tahliyah)
- c) Closely related to leading a successful life.

vi. Worship of Tongue basically means:

- a) Expression of belief through declaration of Shahadah
- b) Calling the people to Islam
- c) Teaching the people recitation of the Quran

vii. One out of different types of 'Ibadah by Organs is:

- a) Performing Hajj and 'Umrah
- b) Giving Sadaqah to the needy people
- c) Making Du'a to Allah (SWT)

viii. The literal meaning of Khilafah is:

- a) Accountability to the Almighty Allah (SWT)
- b) Freedom of will to do anything freely
- c) Succession and Vicegerency

ix. One of the necessary requirements of Khilafah is:

- d) Having the control over the resources of the earth
- e) Becoming the ruler over the creation
- f) Exercising the power of Allah over all creation

x. Allah's declaration for appointing a Khalifa on the earth was:

- a) Related to Adam (AS) only.
- b) Related to Dawud (AS) only.
- c) Related to the Prophet Muhammad (SAW).

Q.[2] Note down the following statements and determine whether they are right or wrong.

- i. Love of someone like the object of worship is also considered as one of the Major Shirk. ✓
- ii. Paying obligatory Zakat and voluntary Sadaqa are also considered as the way to attain purification of soul. ✓
- iii. The soul (nafs) as has been classified in the Quran is of two kinds: e.g. Naf's Ammarah and Nafs Mutma'innah. ✓
- iv. Struggling against the soul (Nafs) is considered a great war. ✓
- v. Purification of souls was the most important task for the Prophet (SAW) after transmission of revelation. ✓

OR

Explain any five foundational principles of Tahliya (التحلية) and discuss how these principles contribute to the refinement of one's character.

Q.[3] Answer the following questions in a sentence:

- i. Give the literal meaning of Tazkiyah.
- ii. What are the two-fold purposes behind the creation of mankind?
- iii. Define 'Ibadah' technically. Does it include social political, economic and administrative dimensions?
- iv. Explain Minor Shirk with example. Why is it considered a kind of Shirk?
- v. "Knowledge and wisdom are crucial in adorning one's soul". Explain.

OR

Identify any five Qur'anic verses and five authentic Hadiths related to Exhortation (الحث والإرشاد) and justify their relevance in motivating believers toward righteous deeds.

PART- [B]: Answer any one of the following questions:

20

Q. [4]: Explain the two fold ways of attaining Tazkiyah with Examples.

Q. [5]: What do you understand by the term the critical appreciation of soul (Muhasabat al-Nafs)? What are the outcomes of one's critical appreciation of soul?

Wishing you the best of success.

International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE Midterm Examination, Autumn 2025

Course Code: STAT 2311 Course Title: Probability and Statistics
Total marks: 50 Time: 2 hour 30 minutes

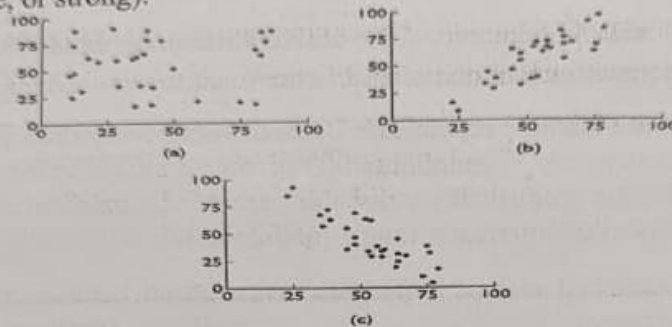
[Answer *all* the questions. Figures in the right-hand margin indicate full marks.
 Separate answer script must be used for Group A and Group B]

Group A

CO DL

- 1. a** Define correlation with suitable examples. Examine each of the scatter plots provided and indicate whether a correlation exists. If a correlation is present, specify its direction (positive or negative) and strength (weak, moderate, or strong).

4 CO2 C3



OR

What is meant by regression analysis? How does it differ from correlation analysis?

4 CO2 C2

- 1. b** A market trader sells ball-point pens on his stall. He sells the pen for a different fixed price (X Taka) in each of six weeks. He notes the number of pens (Y) that he sells in each of these six weeks. The results are shown in the following table.

6 CO2 C3

Week	1	2	3	4	5	6
X	14	18	22	27	32	36
Y	70	64	58	50	39	33

- Calculate the equation of regression line of Y on X.
- Find the number of pens sold by the market trader at a price of 40 Taka in the 7th week.

- 2. a** Define the following terms with examples: random experiment, event, independent event, sample space.

4 CO3 C2

- 2. b** An ensemble system uses three models, M1, M2, and M3, which process 40%, 35%, and 25% of the data respectively, with misclassification probabilities of 5%, 4%, and 6%. If a sample is misclassified, find (i) the probability it was handled by M2 and (ii) the probability it was handled by M1 or M3.

6 CO3 C3

OR

In a neighborhood, 90% children were falling sick due flu and 10% due to measles and no other disease. The probability of observing rashes for measles is 0.95 and for flu is 0.08. If a child develops rashes, find the child's probability of having flu. 6 CO3 C3

Group B

3. a Define a random variable and mention its types. Also, define the probability density function. 4 CO3 C1
3. b A continuous random variable X has the following probability density function: $f(x) = cx^2$; $1 < x < 2$. Find the value of c . Also compute $E(X)$, $E(2X+2)$, $V(X)$, and $V(2X-4)$. 6 CO3 C3
4. a Write the key assumption of a binomial distribution? Mention two applications of the normal distribution in data science. 4 CO3 C2
4. b In a large software system, on average 1 module out of 100 fails during testing. If the system has 300 modules:
(i) What is the probability that at most 2 modules fail?
(ii) What is the probability that at least 1 module fails? 6 CO3 C3
5. a Define hypothesis and test of hypothesis. Distinguish between type-I error and type-II error. 4 CO4 C2

OR

Define null hypothesis and alternative hypothesis with examples. How p value is interpreted? 4 CO4 C2

5. b Write down the steps required to perform a test of hypothesis. The following contingency table shows the classification of 250 peoples according to the gender and Susceptibility with the COVID-19: 6 CO4 C5

COVID-19 Susceptibility	Gender	
	Male	Female
Yes	55	70
No	80	45

Test whether there is any relationship between gender preference and the COVID-19 susceptibility at $\alpha=0.01$. The critical value of chi-square at 1% level of significance with 1 degree of freedom is 6.635.

OR

A company claims that its cloud storage service has an average upload speed of 50 Mbps. A sample of 20 measurements shows a mean speed of 47 Mbps with a standard deviation of 4 Mbps. Test the claim at 1% significance level (two-tailed). The critical value t with 19 degrees of freedom is 2.861. 6 CO4 C5